25901314 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEHISA00 - SCIENCE AND MODERNITY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Explain Isaac Newton's contribution to the development of modern science. [U] / [CO1]
2. Explain the meaning of analytical geometry and its basic principles. [U] / [CO1]
3. Describe Rene Descartes' method of doubt and its importance. [U] / [CO1]
4. Explain the main features of Ptolemy's Geocentric theory of the Universe. [E] / [CO1]
5. What is Cross- Cultural Exchange? [K] / [CO2, CO3]
6. How Gender relates to Witch Hunt? [U] / [CO2, CO3]
7. Define the idea of Mechanical Universe. [U] / [CO2, CO3]
8. Describe Commercial agriculture. [U] / [CO1, CO2]
9. What is Scientic Temper as enshrined in India's Constitution? [U] / [CO3]
10. What is Little Science? [U] / [CO2]
11. Identify the relationship between unity and differentiation in science. [U] / [CO1, CO3]

12. What are the important charcteristics of a social movement. [U] / [CO5]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Analyse the contribution of Jesuit scholars to Early Modern Science. [An] / [CO2]

14. Explain the importance of scientific societies and journals in spreading new scientific ideas and assess their impact on knowledge production. [E] / [CO1, CO2]

15. 30. Discuss the health hazards caused by the Industrial Revolution. [U] / [CO2]

16. Discuss the contributions of Rene Descartes. [U] / [CO2]

17. Apply racial science in administrative policy. [A] / [CO4]

18. How do you understand Michel Foucault’s concept of power? [U] / [CO2]

19. Give a brief account of Samir Amin’s critique of Eurocentrism. [U] / [CO4]

20. Describe the socio-political background of hybrid scientific practices. [U] / [CO3]

21. Explain the meaning of science in your own words. [U] / [CO3]

22. Apply the idea of specialization to explain the emergence of new scientific fields. [A] / [CO4]

[5x6 = 30]

Part C

Essays

Answer any **2** questions

Each question carries **10** marks

23. Evaluate the impact of Galileo's experimental mathematical approach on the scinetific revolution, while explaining its long term influence on later scientists. [E] / [CO1]

24. Examine the role of rationalisation in shaping modern scientific knowledge. [E] / [CO3]

25. Discuss vernacularization of science in colonies. [U] / [CO3]

26. Analyse the historical process behind the formation of scientific disciplines in the nineteenth century. [An] / [CO2]

[10x2 = 20]